

Solutions to Year 2 Quantum Mechanics Unassessed Problem Sheet 4

1. Number of nuclei remaining at time t is $N(t) = N(0) \exp(-t/\tau)$, where τ is the mean lifetime. Definition of half-life, $\tau_{1/2}$, is $N(\tau_{1/2})/N(0) = \exp(-\tau_{1/2}/\tau) = 1/2$. Therefore, $\tau = -\tau_{1/2} / \ln(1/2) = \tau_{1/2} / \ln(2)$.

$$\Delta E/E \geq \hbar / (2\tau E) = \ln(2) \hbar / (2\tau_{1/2} E) = 1.3 \times 10^{-40} \text{ for } ^{232}\text{Th}, \\ = 8.5 \times 10^{-17} \text{ for } ^{212}\text{Po}.$$

2. $\mathbf{L} = \mathbf{r} \wedge \mathbf{p}$

$$L_x = y p_z - z p_y; \quad \hat{L}_x = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right);$$

$$L_y = z p_x - x p_z; \quad \hat{L}_y = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right);$$

$$L_z = x p_y - y p_x; \quad \hat{L}_z = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right).$$

In spherical polar coordinates (r, θ, ϕ) ,

$$x = r \sin \theta \cos \phi \quad \frac{\partial x}{\partial \phi} = -y$$

$$y = r \sin \theta \sin \phi \quad \frac{\partial y}{\partial \phi} = x$$

$$z = r \cos \theta \quad \frac{\partial z}{\partial \phi} = 0$$

$$\text{Now } \frac{\partial}{\partial \phi} = \frac{\partial}{\partial y} \cdot \frac{\partial y}{\partial \phi} + \frac{\partial}{\partial x} \cdot \frac{\partial x}{\partial \phi} + \frac{\partial}{\partial z} \cdot \frac{\partial z}{\partial \phi} \quad (\text{chain rule})$$

$$= \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

$$\therefore -i\hbar \frac{\partial}{\partial \phi} = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) = \hat{L}_z.$$

$$3. \quad \psi(r, \theta, \phi) = R(r) \sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi}$$

$$(a) \quad \hat{L}_z = -i\hbar \frac{\partial}{\partial \phi},$$

$$\hat{L}_z \psi = -i\hbar \frac{\partial}{\partial \phi} \left(R(r) \sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi} \right)$$

$$= -i\hbar \times i \left(R(r) \sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi} \right)$$

$$= \hbar \psi$$

$\therefore \psi(r, \theta, \phi)$ is an eigenfunction of $\hat{L}_z$ with eigenvalue $+\hbar$.

$$(b) \quad \hat{L}^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right),$$

$$\hat{L}^2 \psi = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) \left(R(r) \sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi} \right)$$

$$= -\hbar^2 R(r) \sqrt{\frac{3}{8\pi}} \left[-\sin \theta e^{i\phi} + \cot \theta \cos \theta e^{i\phi} - \frac{1}{\sin^2 \theta} \sin \theta e^{i\phi} \right]$$

$$= -\hbar^2 R(r) \sqrt{\frac{3}{8\pi}} \left[-\sin \theta e^{i\phi} + \frac{\cos^2 \theta}{\sin^2 \theta} \sin \theta e^{i\phi} - \frac{1}{\sin^2 \theta} \sin \theta e^{i\phi} \right]$$

$$= -\hbar^2 R(r) \sqrt{\frac{3}{8\pi}} [-2 \sin \theta e^{i\phi}]$$

$$= 2\hbar^2 \psi$$

$\therefore \psi(r, \theta, \phi)$ is an eigenfunction of $\hat{L}^2$ with eigenvalue $2\hbar^2$.

$$4. \quad L_z = +\hbar, \quad L^2 = 2\hbar^2, \quad \therefore |L| = \sqrt{2}\hbar.$$

Angle between z-axis and vector $\mathbf{L}$ is $\theta = \cos^{-1} (L_z / |L|) = \cos^{-1} (1/\sqrt{2}) = \pi/4$.

$$L^2 = L_x^2 + L_y^2 + L_z^2 = (L_x^2 + L_y^2) + \hbar^2,$$

$$\therefore (L_x^2 + L_y^2) = \hbar^2.$$

The angular momentum in the x - y plane is of magnitude $\hbar$ but of unknown direction. If the direction of the angular momentum in this plane were known precisely, then the direction of $\mathbf{L}$ would be known precisely. This is not allowed by the Uncertainty Principle. Therefore the separate components L_x and L_y cannot be known.

5. $L = r p_t,$

(a) $\Delta s = \Delta \phi r,$

$$\Delta p_t = \Delta L / r,$$

$$\therefore \Delta s \Delta p_t = \Delta \phi r \Delta L / r = \Delta \phi \Delta L .$$

$$\therefore \text{ If } \Delta s \Delta p_t \geq \hbar / 2 , \text{ then } \Delta \phi \Delta L \geq \hbar / 2 .$$

(b) In a Bohr orbit the magnitude of L is known precisely, because the wavefunction is an eigenfunction of $\hat{L}^2$. Therefore we can have no information on the ϕ position of the electron.

(c) $\Delta \phi = (1/6)$ radians and $\Delta \phi \Delta L \geq \hbar / 2 .$

$$\therefore \Delta L \geq \frac{\hbar / 2}{\Delta \phi} = \frac{\hbar / 2}{1/6} = 3\hbar .$$

$\therefore$ Eigenfunctions covering a range of $|L|$ values of at least $3\hbar$ must be present in the superposition e.g. $\ell = 1, 2, 3$ and 4 .

6. Spin $\pm 1/2 \hbar$ in the z direction.

Magnetic moment of the atom $\mu = m_s g_s \mu_B .$

Bohr magneton, $\mu_B = 0.927 \times 10^{-23} \text{ amp-m}^2 , \quad m_s = \pm 1/2, \quad g_s = 2.00 .$

$$\text{Force in } z \text{ direction} = \mu \frac{\partial B}{\partial z} .$$

The acceleration for an atom of mass m is given by

$$\mu \frac{\partial B}{\partial z} = ma .$$

$$\text{Thermal energy of atoms} = \frac{3}{2} kT = \frac{1}{2} mv^2 ,$$

$$\therefore \text{ velocity of atoms, } v = \sqrt{\frac{3kT}{m}} .$$

$$\therefore \text{Time spent in magnet} = \frac{0.5}{\sqrt{\frac{3kT}{m}}}.$$

$$\begin{aligned} \text{Using } s &= ut + \frac{1}{2}at^2 \text{ we get a displacement in magnet} = \frac{1}{2}at^2 \\ &= \frac{1}{2} \times \mu \frac{\partial B}{\partial z} \times \frac{0.25m}{3kT} \\ &\quad \frac{1}{m} \end{aligned}$$

$$= \frac{0.5 \times 0.927 \times 10^{-23} \times 10 \times 0.25}{3 \times 1.38 \times 10^{-23} \times 500}$$

$$= 5.6 \times 10^{-4} \text{ m.}$$

Transverse velocity developed in magnet ($v = u + at$)

$$= \mu \frac{\partial B}{\partial z} \frac{0.5 \sqrt{m}}{\sqrt{3kT} m}$$

$$= \frac{0.927 \times 10^{-23} \times 10 \times 0.5}{\sqrt{3 \times 1.38 \times 10^{-23} \times 500 \times 1.67 \times 10^{-27}}}$$

$$= 7.9 \text{ m/sec.}$$

$$\begin{aligned} \text{Longitudinal velocity} &= \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 500}{1.67 \times 10^{-27}}} \\ &= 3521 \text{ m/sec.} \end{aligned}$$

$$\text{At 2m, deflection} = \frac{2 \times 7.9}{3521} = 4.5 \times 10^{-3} \text{ m.}$$

$$\begin{aligned} \therefore \text{Total deflection} &= 4.5 \times 10^{-3} + 5.6 \times 10^{-4} \\ &= 5.06 \times 10^{-3} \text{ m.} \end{aligned}$$

$$\therefore \text{Separation of images} = \underline{10.12 \text{ mm.}}$$